

# NBBO

February 13, 2026

## 1 Question 3

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

### 1.1 Part a)

Stock selection and data import. I selected the Moderna (MRNA) stock, which is a top 1000-sized stock (beforehands, I tested my code on the italian company Trivago (TRVG), which is much smaller).

```
[2]: taq = pd.read_csv('data/TAQ_MRNA_quotes_sep2025.csv')
```

Filter data:

```
[3]: import sys
sys.path.append('.')
from utils import filter_data_by_market_hours
```

```
[4]: tickers = taq['SYM_ROOT'].unique()
print(tickers)
```

<StringArray>

['MRNA']

Length: 1, dtype: str

```
[ ]: taq_filtered = filter_data_by_market_hours(taq)

# print(taq_filtered.tail(20))

grouped = taq_filtered.groupby('DATE')

# print(grouped.head())
```

|          | DATE       | TIME_M          | EX  | BID | BIDSIZ | ASK | ASKSIZ | \  |
|----------|------------|-----------------|-----|-----|--------|-----|--------|----|
| 19754431 | 30/09/2025 | 16:10:48.348517 | 440 | Z   | 25.77  | 5   | 29.78  | 5  |
| 19754432 | 30/09/2025 | 16:10:48.348579 | 059 | Z   | 25.77  | 7   | 29.78  | 5  |
| 19754433 | 30/09/2025 | 16:10:48.348600 | 143 | Y   | 25.77  | 1   | 39.95  | 28 |

|          |            |                    |   |       |    |       |    |
|----------|------------|--------------------|---|-------|----|-------|----|
| 19754434 | 30/09/2025 | 16:10:48.348606278 | J | 25.77 | 1  | 27.16 | 1  |
| 19754435 | 30/09/2025 | 16:10:48.348607203 | K | 25.77 | 1  | 25.98 | 2  |
| 19754436 | 30/09/2025 | 16:10:48.348874527 | P | 25.77 | 1  | 25.81 | 1  |
| 19754437 | 30/09/2025 | 16:10:48.356507445 | K | 25.30 | 1  | 25.98 | 2  |
| 19754438 | 30/09/2025 | 16:10:48.357258320 | Q | 25.78 | 20 | 25.91 | 1  |
| 19754439 | 30/09/2025 | 16:10:48.366686745 | Z | 25.77 | 5  | 29.78 | 5  |
| 19754440 | 30/09/2025 | 16:10:48.367432083 | P | 25.68 | 1  | 25.81 | 1  |
| 19754441 | 30/09/2025 | 16:10:48.374402464 | J | 24.62 | 1  | 27.16 | 1  |
| 19754442 | 30/09/2025 | 16:10:48.588006678 | Y | 0.00  | 0  | 39.95 | 28 |
| 19754443 | 30/09/2025 | 16:10:50.534072936 | K | 25.30 | 1  | 25.98 | 2  |
| 19754444 | 30/09/2025 | 16:11:25.734677883 | Z | 25.77 | 5  | 29.78 | 5  |
| 19754445 | 30/09/2025 | 16:13:08.217195561 | U | 25.50 | 1  | 26.05 | 1  |
| 19754446 | 30/09/2025 | 16:13:12.672241975 | Z | 25.77 | 5  | 26.00 | 1  |
| 19754447 | 30/09/2025 | 16:14:25.086590891 | U | 25.50 | 1  | 26.05 | 1  |
| 19754448 | 30/09/2025 | 16:14:45.136254305 | V | 0.00  | 0  | 25.80 | 1  |
| 19754449 | 30/09/2025 | 16:14:56.249356683 | V | 0.00  | 0  | 0.00  | 0  |
| 19754450 | 30/09/2025 | 16:14:56.250034577 | Q | 25.78 | 17 | 25.91 | 1  |

|          | NATBBO_IND | SYM_ROOT | SYM_SUFFIX |        | timedelta          |
|----------|------------|----------|------------|--------|--------------------|
| 19754431 | 0          | MRNA     | NaN        | 0 days | 16:10:48.348517440 |
| 19754432 | 2          | MRNA     | NaN        | 0 days | 16:10:48.348579059 |
| 19754433 | 0          | MRNA     | NaN        | 0 days | 16:10:48.348600143 |
| 19754434 | 0          | MRNA     | NaN        | 0 days | 16:10:48.348606278 |
| 19754435 | 0          | MRNA     | NaN        | 0 days | 16:10:48.348607203 |
| 19754436 | 0          | MRNA     | NaN        | 0 days | 16:10:48.348874527 |
| 19754437 | 0          | MRNA     | NaN        | 0 days | 16:10:48.356507445 |
| 19754438 | 2          | MRNA     | NaN        | 0 days | 16:10:48.357258320 |
| 19754439 | 0          | MRNA     | NaN        | 0 days | 16:10:48.366686745 |
| 19754440 | 0          | MRNA     | NaN        | 0 days | 16:10:48.367432083 |
| 19754441 | 0          | MRNA     | NaN        | 0 days | 16:10:48.374402464 |
| 19754442 | 0          | MRNA     | NaN        | 0 days | 16:10:48.588006678 |
| 19754443 | 0          | MRNA     | NaN        | 0 days | 16:10:50.534072936 |
| 19754444 | 0          | MRNA     | NaN        | 0 days | 16:11:25.734677883 |
| 19754445 | 0          | MRNA     | NaN        | 0 days | 16:13:08.217195561 |
| 19754446 | 0          | MRNA     | NaN        | 0 days | 16:13:12.672241975 |
| 19754447 | 0          | MRNA     | NaN        | 0 days | 16:14:25.086590891 |
| 19754448 | 2          | MRNA     | NaN        | 0 days | 16:14:45.136254305 |
| 19754449 | 2          | MRNA     | NaN        | 0 days | 16:14:56.249356683 |
| 19754450 | 2          | MRNA     | NaN        | 0 days | 16:14:56.250034577 |

|          | DATE       | TIME_M            | EX  | BID   | BIDSIZ | ASK   | ASKSIZ | \ |
|----------|------------|-------------------|-----|-------|--------|-------|--------|---|
| 8584     | 02/09/2025 | 9:30:00.044298101 | N   | 22.73 | 1      | 24.23 | 3      |   |
| 8585     | 02/09/2025 | 9:30:00.044303676 | N   | 23.64 | 3      | 24.23 | 3      |   |
| 8586     | 02/09/2025 | 9:30:00.088882012 | Q   | 23.91 | 1      | 23.94 | 1      |   |
| 8587     | 02/09/2025 | 9:30:00.089058955 | Q   | 23.91 | 1      | 24.12 | 3      |   |
| 8588     | 02/09/2025 | 9:30:00.089299477 | Q   | 23.92 | 1      | 24.12 | 3      |   |
| ...      | ...        | ...               | ... | ...   | ...    | ...   | ...    |   |
| 18772362 | 30/09/2025 | 9:30:00.014073080 | V   | 23.78 | 1      | 26.36 | 1      |   |
| 18772363 | 30/09/2025 | 9:30:00.043317330 | N   | 23.78 | 1      | 25.40 | 3      |   |

|          |            |                   |   |       |   |       |     |
|----------|------------|-------------------|---|-------|---|-------|-----|
| 18772364 | 30/09/2025 | 9:30:00.043318343 | N | 24.78 | 3 | 25.40 | 3   |
| 18772365 | 30/09/2025 | 9:30:00.061340841 | P | 25.04 | 1 | 25.13 | 3   |
| 18772366 | 30/09/2025 | 9:30:00.158715562 | Q | 25.04 | 7 | 25.10 | 379 |

|          | NATBBO_IND | SYM_ROOT | SYM_SUFFIX | timedelta                 |
|----------|------------|----------|------------|---------------------------|
| 8584     | 0          | MRNA     | NaN        | 0 days 09:30:00.044298101 |
| 8585     | 0          | MRNA     | NaN        | 0 days 09:30:00.044303676 |
| 8586     | 2          | MRNA     | NaN        | 0 days 09:30:00.088882012 |
| 8587     | 2          | MRNA     | NaN        | 0 days 09:30:00.089058955 |
| 8588     | 0          | MRNA     | NaN        | 0 days 09:30:00.089299477 |
| ...      | ...        | ...      | ...        | ...                       |
| 18772362 | 0          | MRNA     | NaN        | 0 days 09:30:00.014073080 |
| 18772363 | 0          | MRNA     | NaN        | 0 days 09:30:00.043317330 |
| 18772364 | 0          | MRNA     | NaN        | 0 days 09:30:00.043318343 |
| 18772365 | 0          | MRNA     | NaN        | 0 days 09:30:00.061340841 |
| 18772366 | 4          | MRNA     | NaN        | 0 days 09:30:00.158715562 |

[105 rows x 11 columns]

```
[22]: start_time = pd.Timedelta(hours=9, minutes=30)
end_time = pd.Timedelta(hours=16)
grid_times = pd.timedelta_range(start=start_time, end=end_time, freq='1min')

midpoints_by_day = {}

for date, group in grouped:
    group = group.sort_values('timedelta')
    # Filter out invalid quotes (BID <=0 or ASK <=0)
    group = group[(group['BID'] > 0) & (group['ASK'] > 0)]
    # Filter out bad quotes where spread is unreasonably large (> 1% of bid_
    ↪price)
    group = group[(group['ASK'] - group['BID']) / group['BID'] < 0.01]

    midpoints = []

    for grid_time in grid_times:
        before_quotes = group[group['timedelta'] <= grid_time]
        if not before_quotes.empty:
            last_quote = before_quotes.iloc[-1]
            midpoint = (last_quote['BID'] + last_quote['ASK']) / 2
        else:
            # For 09:30, if no before, use first quote of the day
            if grid_time == start_time:
                first_quote = group.iloc[0] if not group.empty else None
                if first_quote is not None:
                    midpoint = (first_quote['BID'] + first_quote['ASK']) / 2
            else:
```

```

        midpoint = float('nan')
    else:
        # No quote before, set to NaN
        midpoint = float('nan')
    midpoints.append(midpoint)

midpoints_series = pd.Series(midpoints, index=grid_times)
midpoints_by_day[date] = midpoints_series

```

Debugging cell:

```

[ ]: # MDRNA_midpoints = midpoints_by_day
# print(midpoints_by_day['02/09/2025'].loc[:])
# print(midpoints_by_day['02/09/2025'].to_numpy())
# print(midpoints_by_day['02/09/2025'].index.to_numpy())

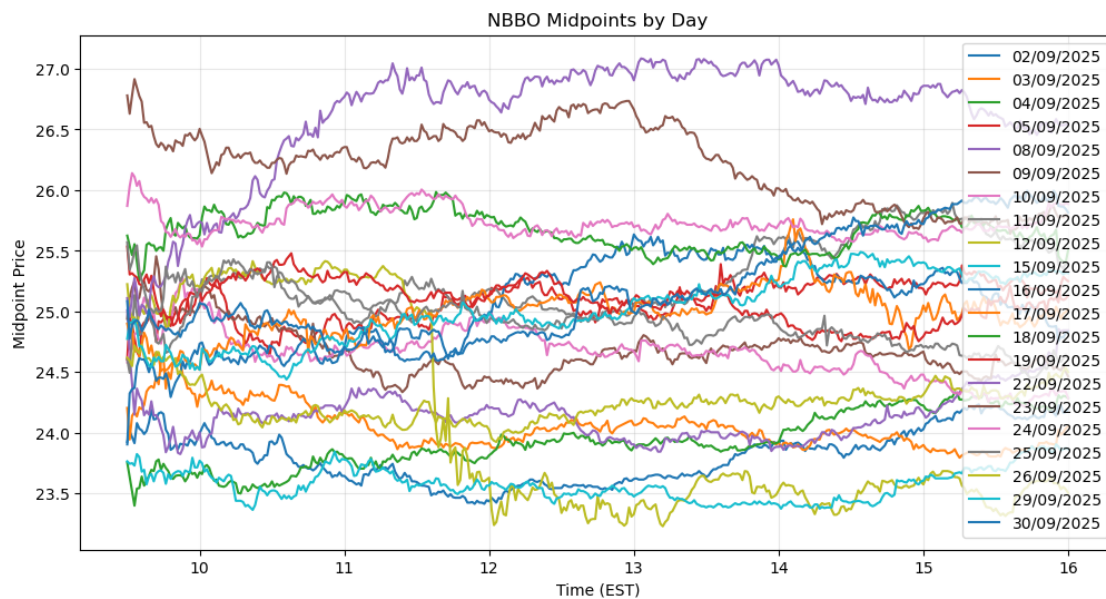
```

```

[26]: plt.figure(figsize=(12, 6))
for day, midpoints in midpoints_by_day.items():
    x_vals = midpoints.index.total_seconds() / 3600
    plt.plot(x_vals, midpoints, label = day)

plt.xlabel('Time (EST)')
plt.ylabel('Midpoint Price')
plt.legend()
plt.title('NBBO Midpoints by Day')
plt.grid(True, alpha=0.3)
plt.show()

```



## 1.2 Part c)

Compute price differences and intraday variance curve

```
[ ]: # print(midpoints_by_day['02/09/2025'].diff().dropna())

bin_duration = pd.Timedelta(minutes=15)
bins = []
current = start_time
while current < end_time:
    bins.append(current)
    current += bin_duration

variance_curve = {}
for bin_start in bins:
    bin_end = bin_start + bin_duration
    sum_sq_list = []
    for date, midpoints in midpoints_by_day.items():
        in_bin = (midpoints.index >= bin_start) & (midpoints.index < bin_end)
        midpoints_in_bin = midpoints.loc[in_bin]
        if len(midpoints_in_bin) > 1:
            diffs_in_bin = midpoints_in_bin.diff().dropna()
            sum_sq = (diffs_in_bin ** 2).sum()
            sum_sq_list.append(sum_sq)
    if sum_sq_list:
        avg_variance = np.mean(sum_sq_list)
    else:
        avg_variance = 0
    variance_curve[bin_start] = avg_variance

variance_series = pd.Series(variance_curve)

plt.figure(figsize=(12, 6))
x_vals = variance_series.index.total_seconds() / 3600
plt.plot(x_vals, variance_series.values, marker='o')
plt.xlabel('Time (EST)')
plt.ylabel('Average Intraday Variance')
plt.title('15-Minute Intraday Variance Curve for MRNA')
plt.grid(True, alpha=0.3)
plt.show()
```

